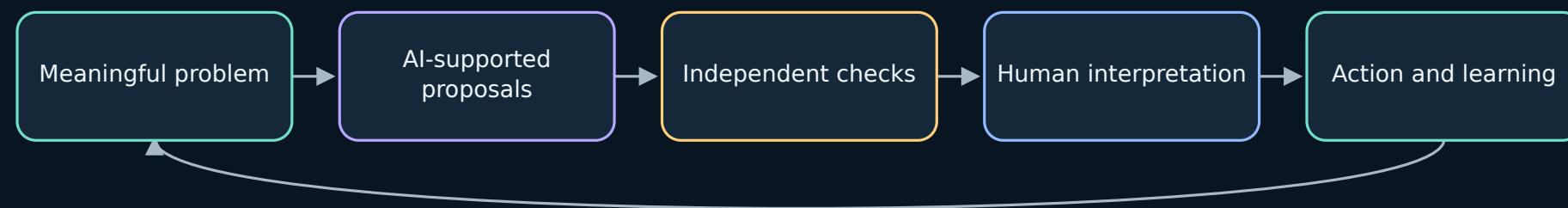


What changes when AI can help us discover?

当 AI 开始参与发现，我们的价值在哪里？

- AI is entering parts of the research process.
- The opportunity is broader than faster writing.
- Our role: define worthwhile problems and establish trustworthy answers.

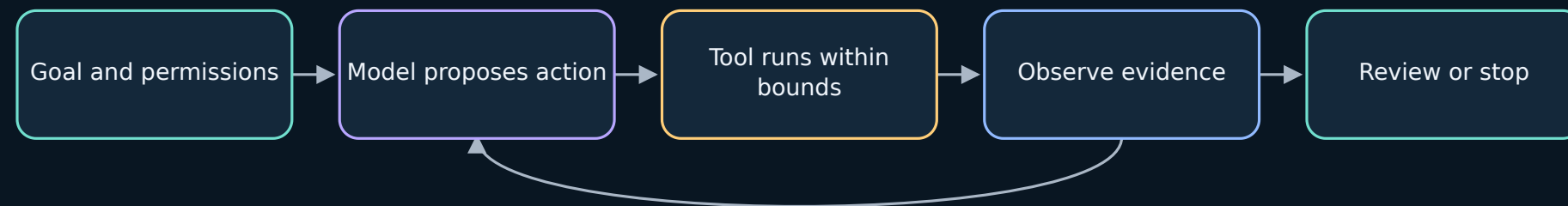


Conceptual workflow: proposals become useful through evidence and accountable decisions.

A model becomes useful through a system

模型能力要通过工具、数据和验证转化为价值。

- LLM: generates language and other learned representations.
- Reasoning workflow: spends effort exploring and checking steps.
- Agent: uses tools, observes results and chooses further actions.
- A fluent explanation does not establish correctness.

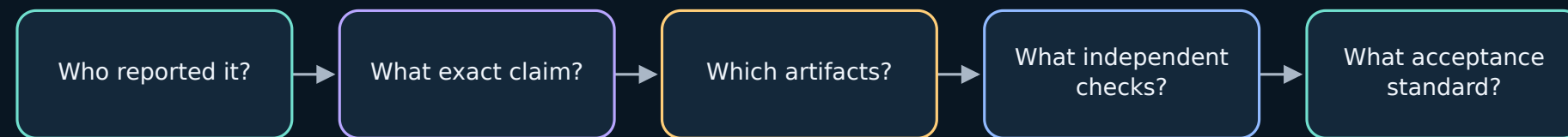


Tool access and a stopping condition are explicit parts of the system.

Read the evidence before repeating the headline

先判断证据，再传播结论。

- Ask who made the claim and when.
- Read the exact problem and assumptions.
- Locate the paper, code, data or certificate.
- Separate artifact checking, scientific acceptance and practical usefulness.

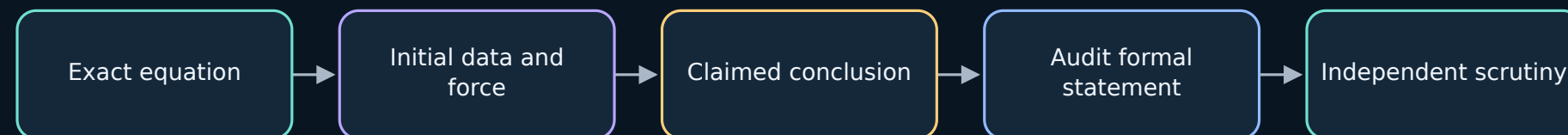


Different disciplines use different acceptance standards. These are questions, not a universal linear ranking.

Navier–Stokes: a precise, attributed frontier claim

重大数学进展，要精确表述并接受独立检验。

- OpenAI announced a proposed solution on September 8, 2026.
- The manuscript constructs finite-time blowup with smooth forcing and positive viscosity.
- The exact forced formulation matters.
- Independent acceptance and prize recognition are separate.

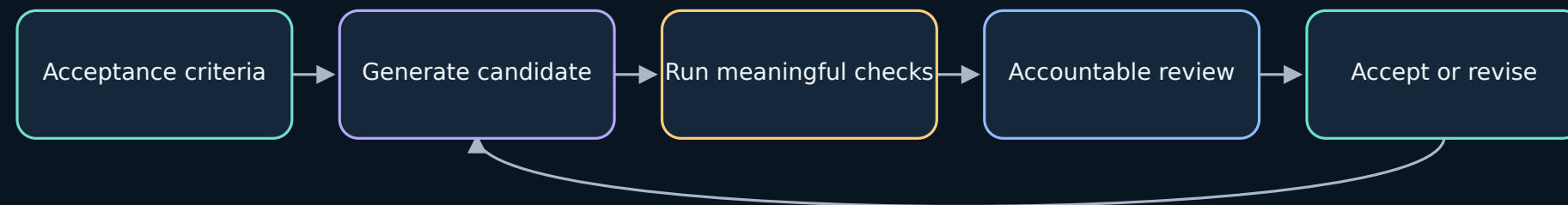


Interpretation guide; this package does not certify a mathematical proof.

Verification is infrastructure for discovery

把验证建设成基础设施。

- Specify what a valid answer must satisfy.
- Use checks that can reject the proposal.
- Keep provenance, assumptions and failure cases.
- Spend verification effort where errors matter most.

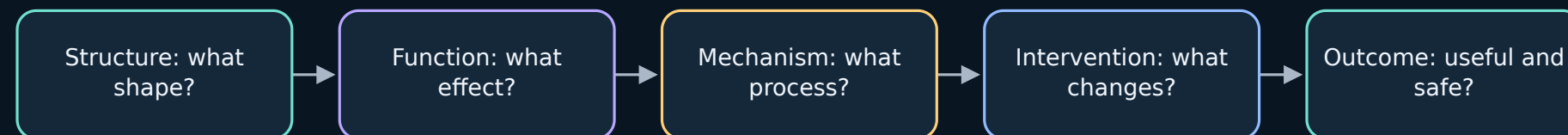


Use checks capable of rejecting a generated result; select the reviewer and criteria before acting.

Scientific AI is a broader ecosystem than LLMs

科学 AI 不只有大语言模型。

- AlphaFold 3 predicts structures of biomolecular complexes.
- Specialized predictors and generators support different tasks.
- LLMs can help connect literature, code and tools.
- Structure, function, mechanism and clinical benefit require different evidence.

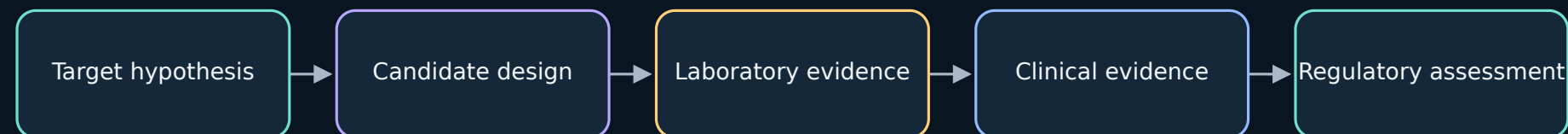


A shape prediction does not by itself identify function, mechanism or therapeutic benefit.

Drug discovery still ends in evidence from people

候选分子不等于获批药物，临床证据不可跳过。

- AI can support target selection and candidate design.
- Rentosertib has a published phase 2a IPF trial.
- That study involved 71 patients and a 12-week treatment period.
- Its primary objective was safety and tolerability; broader benefit needs further evidence.

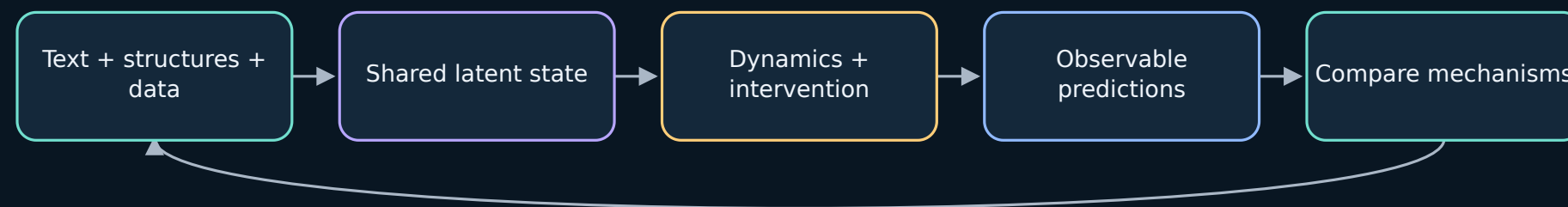


A simplified educational pipeline; stages and regulatory requirements differ by product and jurisdiction.

From prediction and generation to testable mechanisms

从预测与生成，走向可检验的机制假说。

- Repository contribution: Shengchao Liu' s Wave Intelligence perspective.
- Connect text, structures and measurements through shared representations.
- Model how a system evolves under conditions and interventions.
- Compare alternative mechanisms using observable predictions.

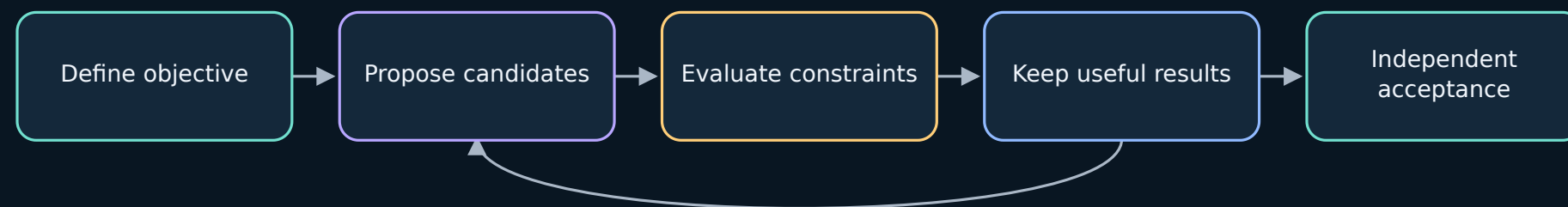


Original explanatory adaptation of Liu' s repository perspective, pp. 3–6; a research agenda, not an established validated system.

Research agents need feedback from the world

研究智能体需要来自真实世界的反馈。

- Coscientist demonstrated tool-connected laboratory workflows.
- AlphaEvolve connects program proposals to automated evaluation.
- The evaluator determines what progress means.
- Simulation success needs an appropriate bridge to reality.

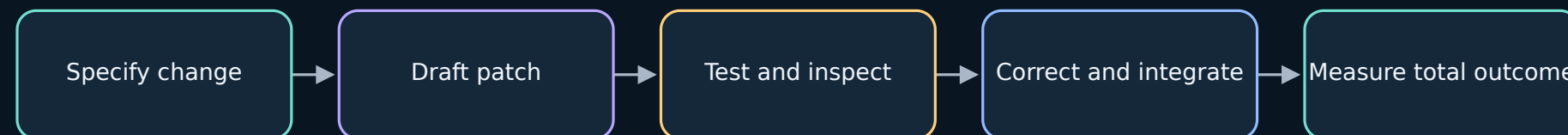


Conceptual abstraction of tool-connected discovery; simulation and physical validation can have different requirements.

In software, measure the whole task

衡量整个任务，而不是生成代码的速度。

- AI can assist drafting, debugging, tests and code exploration.
- METR' s early-2025 experiment found 19% longer task time in its setting.
- The February 2026 follow-up identifies major measurement biases.
- Measure current tools on your own tasks, including review and rework.

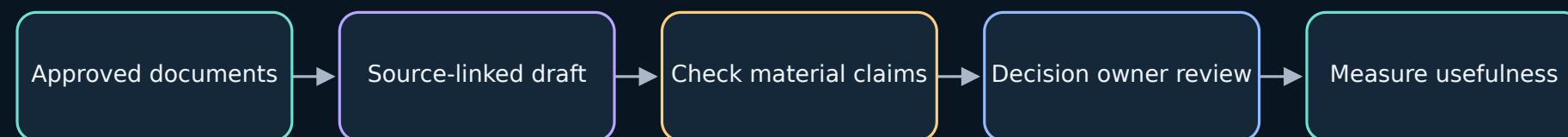


Generation time is only one component. Acceptance includes review and integration.

Knowledge work depends on context and quality

知识工作的收益取决于场景和质量。

- A support-agent study found heterogeneous productivity gains.
- Its November 2024 manuscript version reports 15% on average.
- That metric is issues resolved per hour in one work setting.
- Transfer requires access to relevant knowledge and quality checks.

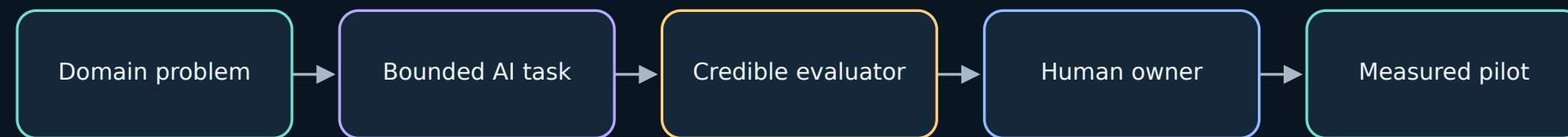


The final decision owner checks material claims and the completeness of the evidence.

Find the opportunity inside your discipline

机会来自专业知识与 AI 能力的结合。

- Engineering: explore alternatives; check constraints and simulation validity.
- Science: organize evidence; design discriminating experiments.
- Software: propose changes; measure integration quality.
- Leadership: redesign workflows; retain accountable owners.

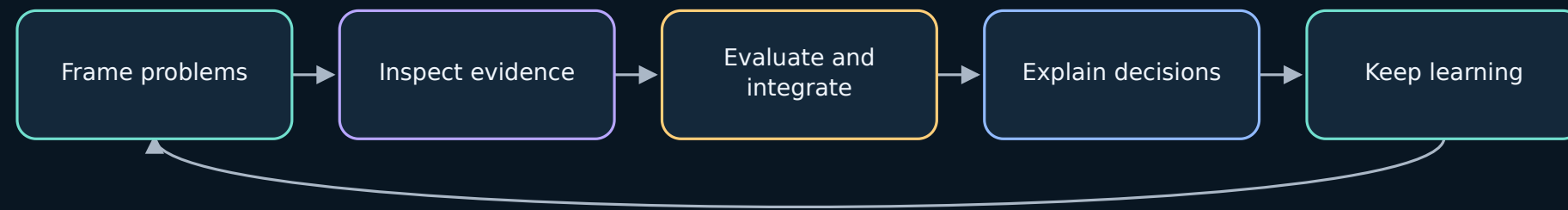


Illustrative suggestions for alumni; no claims about an existing DUT alumni program.

Build the skills that make assistance dependable

会提问，更要会验证、会整合、会负责。

- Problem formulation and domain judgment.
- Data literacy, provenance and uncertainty.
- Evaluation, experimentation and tool integration.
- Communication, accountability and continued learning.

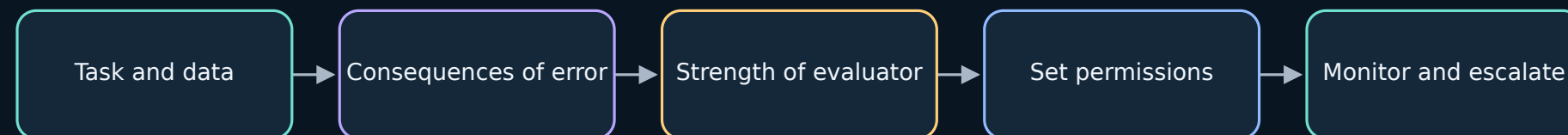


Recommended capabilities to practice together, rather than a forecast of job counts.

Match autonomy to consequences

自主程度要与后果和验证能力相匹配。

- Confident errors and unsupported citations.
- Sensitive data and malicious instructions in retrieved content.
- Unequal outcomes, deskilling and unclear responsibility.
- Use bounded permissions, evidence checks and accountable review.

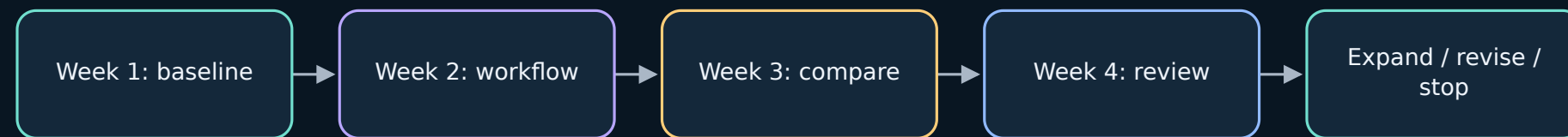


Suggested decision aid, not a numeric risk model or legal determination.

Run one measured pilot in the next 30 days

30 天内做一个可衡量、可复盘的试点。

- Week 1: choose a task, owner and baseline.
- Week 2: define inputs, checks and bounded access.
- Week 3: compare matched tasks and record failures.
- Week 4: decide to expand, revise or stop.

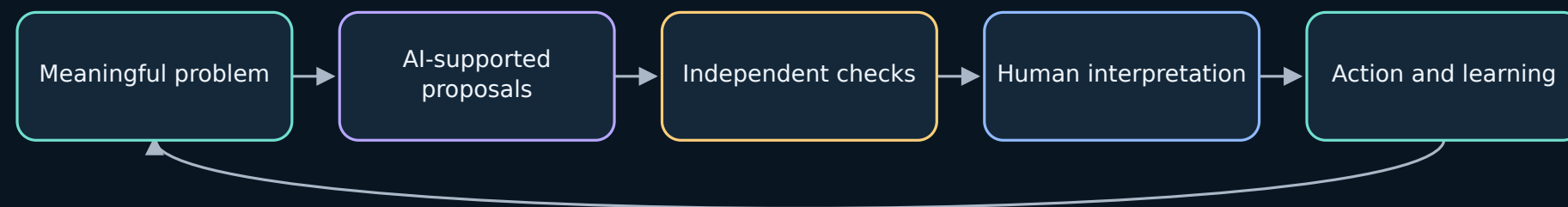


Recommended schedule. Expand, revise and stop are all legitimate outcomes.

More ideas. Stronger evidence. Human responsibility.

让 AI 扩展探索，让证据决定行动，让人承担责任。

- Use AI to widen the space of possibilities.
- Use evidence to decide what deserves action.
- Build your career around judgment and reliable execution.
- Panel question: which alumni problem has both value and a credible evaluator?



Conceptual workflow: proposals become useful through evidence and accountable decisions.